Reinforcement Learning for Equitable NYC Housing Inspections

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This presentation applies machine learning to a practical problem in NYC housing enforcement: helping the Department of Buildings decide which 311 complaints are actually worth sending an inspector to. Using real NYC 311 data and reinforcement learning — a technique where a computer learns from experience, like a dispatcher who gets better at triage over time — this project builds and tests an automated complaint prioritization system designed to be both efficient and fair across all five boroughs.

RESEARCH FOCUS

- Machine Learning applied to NYC housing code enforcement
- After-Hours Variance (AHV) complaint triage for the Department of Buildings
- Multi-objective optimization: balancing enforcement speed, cost, and tenant protection
- Fairness constraints to prevent algorithmic over-policing of specific boroughs

Benefits to the NYC Department of Buildings

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THE PROBLEM TODAY

Every year, New Yorkers file thousands of 311 complaints about illegal after-hours construction (AHV complaints). The vast majority turn out not to be real violations — meaning inspectors are routinely dispatched to buildings that don't need enforcement action.

97.9% of AHV complaints are NOT real violations

739/755 inspections wasted under an inspect-everything policy

⚠ **Fairness Risk** Without triage, enforcement inadvertently concentrates in certain boroughs

WHAT REINFORCEMENT LEARNING DELIVERS

A learning-based triage system trains on complaint data and figures out which signals actually predict real violations. The result:

100% violation detection — 16 out of 16 found

94% reduction in wasted inspections (739 → 41)*

Built-in Fairness Model is penalized if >40% of inspections concentrate in one borough

Fully Explainable Every dispatch decision traceable to specific complaint features — no black box

** 94% reduction compares the naive Always-Inspect baseline (739 wasted inspections) to the Min-Cost Pareto-optimal policy (41 wasted).*

Data: NYC 311 After-Hours Variance Complaints

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755	16	5	3
AHV Complaint Records	Confirmed Violations (2.1%)	Input Features Per Complaint	Possible Actions Per Decision

⚠ **Data Transparency Note:** The public NYC 311 dataset does not include confirmed violation outcomes — that information exists only in DOB's internal systems. Violation labels were synthetically generated to match the DOB's known ~2% AHV violation rate. The 311 complaint records themselves are real.

THE 5 INPUT FEATURES	
1. Complaint Type	Brand-new / Recurrent / High-frequency building
2. Borough	Which of the 5 NYC boroughs
3. Inspector Budget	How many inspectors are available right now
4. Prior Complaint Count	How many times this building has been flagged before
5. Time of Day	Night, morning, afternoon, or evening complaint

Buildings with repeated complaint histories and complaints filed late at night are the strongest predictors of real violations.

HOW THE MODEL SCORES DECISIONS

The model is designed so that missing a real violation is the worst possible outcome — penalized 50× more heavily than a wasted inspection.

This forces the system to prioritize catching every violation over efficiency, consistent with DOB's mandate to protect tenants.

Methodology: How the System Learns

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What is Reinforcement Learning? A type of machine learning where a system learns by trial and error — like a new dispatcher who gets feedback after every decision and gradually builds intuition about which complaints are worth acting on. The system observes each complaint, chooses an action, and receives a score. Over thousands of iterations, it learns the patterns that distinguish real violations from noise.

INPUT

5 complaint features:
type, borough, budget, history, time

DECISION

Dismiss / Standard Inspection / Aggressive Enforcement

SCORE

Based on whether a real violation actually existed

APPROACHES TESTED

✓ **Tabular Q-Learning — WINNER —** Learned precise, interpretable rules. Only approach that achieved full violation detection. Works because it maintains separate estimates for every possible complaint combination — not fooled by the rare-violation imbalance.

✗ **Neural Networks (DQN & REINFORCE) — Failed —** Both defaulted to dismissing every complaint. Root cause: with only 2.1% real violations, the neural network learned that 'always dismiss' was an easy way to score well. A known failure mode on highly imbalanced data.

– **Monte Carlo — Partial —** Caught 14 of 16 violations but required more inspector dispatches than the winning approach.

Multi-Objective Analysis: 12 versions of the model were tested, each prioritizing a different balance of three goals: catching violations quickly, minimizing wasted inspections, and avoiding aggressive enforcement orders that could displace tenants. The results form a Pareto frontier — a set of configurations where improving one goal requires accepting a tradeoff on another.

Key Findings & Policy Recommendation

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ALGORITHM PERFORMANCE

Approach	Violations Found	Wasted	Outcome
TD Q-Learning ★	**16/16 (100%)**	257	**✓ Winner**
Monte Carlo	14/16 (88%)	594	Partial
Neural Net (DQN)	0/16 (0%)	0 (dismissed all)	**✗ Failed**
REINFORCE	0/16 (0%)	0 (dismissed all)	**✗ Failed**
Random Baseline	12/16 (75%)	464	Inconsistent
Always-Inspect	16/16 (100%)	**739**	Expensive

Neural networks failed because the 2.1% violation rate made dismissal the path of least resistance. Q-Learning succeeded by maintaining independent estimates for each complaint pattern.

POLICY-RELEVANT CONFIGURATIONS (PARETO FRONTIER)

Min-Cost: All 16 violations caught · 41 wasted inspections · 6 housing units at risk

Max-Retention: All 16 violations caught · 45 wasted inspections · Only 2 units at risk — best for tenant protection

Conservative: 23 wasted inspections (lowest cost) · Accepts missing 3 violations · 3 units at risk

WHAT THE MODEL LEARNED (SHAPLEY VALUES)

Shapley values = how much each data signal drives decisions — making the system transparent and auditable.

#1 **Inspector Budget (15.55)** — Tight staffing forces dismissal regardless of complaint type

#2 **Prior Complaints (10.11)** — Repeat-offender buildings best predict real violations

#3 **Time of Day (8.10)** — After-hours night complaints are disproportionately genuine

#4 **Complaint Category (7.51)** — High-frequency area complaints carry stronger signal

#5 **Borough (6.62)** — Geographic enforcement patterns are learned and tracked

POLICY RECOMMENDATION

Deploy the Min-Cost or Max-Retention configuration as an AI triage layer for incoming AHV complaints. The model flags only high-probability cases for inspector dispatch — maintaining 100% violation detection at a fraction of current inspection volume.

The choice is a policy decision, not a technical one: How much inspector cost is acceptable? How much tenant displacement is tolerable? This system makes that tradeoff explicit and adjustable.